



Purple Team

FA2026 • 2026-09-01

Intro to Purple Team

Ronan Boyarski

Ronan Boyarski

- Purple Team Offensive Cyber Lead
- MSCS Student
- Former competitive shooter



Ronan Boyarski

- Purple Team Offensive Cyber Lead
- MSCS Student
- Former competitive shooter
- Also I have a pig



VM Setup



Download VMWare

- Go here:

<https://www.vmware.com/products/desktop-hypervisor/workstation-and-fusion>

- You may need to create a broadcom account
- Search for VMWare Workstation Pro on Windows
- Mac needs VMWare Fusion or UTM depending on Intel or ARM



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Download VMWare - Windows

← VMware Workstation Pro (For Windows) 26H1

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VMware Workstation Pro (For Windows)

Release 26H1

Release Level Info 543221

File Name	Release Date	Last Updated	SHA2	MD5	
VMware Workstation Pro for Windows VMware-Workstation-Full-26H1-25388281.exe(274.34 MB) Build Number: 25388281	May 14, 2026	May 05, 2026	a0ef9087607d9cad20b08139e73e41242e044ad5bd8cee141d3bad314586737f	83f9b6cc7cf0e74ad3b15ccca27860af	

1 to 1 of 1 records

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Download VMWare - Intel Mac

← VMware Fusion (for Intel-based and Apple silicon Macs) 26H1

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543219

English

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VMware Fusion (for Intel-based and Apple silicon Macs)

Release 26H1

Release Level Info 543219

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File Name	Release Date	Last Updated	SHA2	MD5	
VMware Fusion (for Intel-based and Apple silicon Macs) VMware-Fusion-26H1-25388279_universal.dmg(480.71 MB) Build Number: 25388279	May 14, 2026	May 05, 2026	c1d373aa21be25674e3ecc518819e255785dea9d456d8747bcb0a2a59244bdf6	42414933dd368abd622b15e70ceff8a1	

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Kali Install - Not ARM Mac

- Go here: <https://www.kali.org/get-kali/#kali-platforms>
- You want this:



Virtual Machines

- ✓ Snapshots functionary
- ✓ Isolated environment
- ✓ Customized Kali kernel
- ✗ Limited direct access to hardware
- ✗ Higher system requirements

VMware & VirtualBox pre-built images. Allowing for a Kali install without altering the host OS with additional features such as snapshots. Vagrant images for quick spin-up also available.

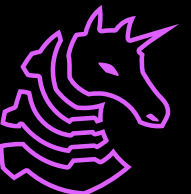
Recommended

Recommended



VMware

3.6G ↓	torrent	docs	sum
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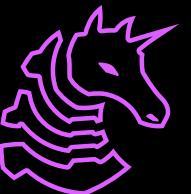
Download VMWare & Kali - ARM Mac

- This will require UTM (<https://mac.getutm.app/>)
- Official docs:
<https://www.kali.org/docs/virtualization/install-utm-guest-vm/>
- I believe this requires the Apple Silicon installer
<https://www.kali.org/get-kali/#kali-installer-images>



Reasonable Defaults

- Create a new VM -> Kali ISO -> Customize Hardware
- You will probably want 60-80 gigs of disk space
- Multiple cores & >4gb RAM preferable
- There should be a guided graphical install process
- Do not change things like the window manager unless you know what you're doing
- If you go too low on disk space you might need to reinstall which is no fun

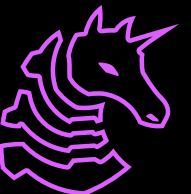


What is Purple Team About?



General Philosophy

- Mix of practical offensive and defensive security topics
- Defenders will understand what advanced attackers actually do
- Attackers will understand high-end defenses
- Some prerequisite knowledge is encouraged
 - Fundamentals in security, scripting, networking, and operating systems
- SIGPwny general teaches fundamentals of security
- We teach how to **actually** attack targets (end-to-end) and defend endpoints within IT/OT networks
 - Configure, build, target, exploit, and report!



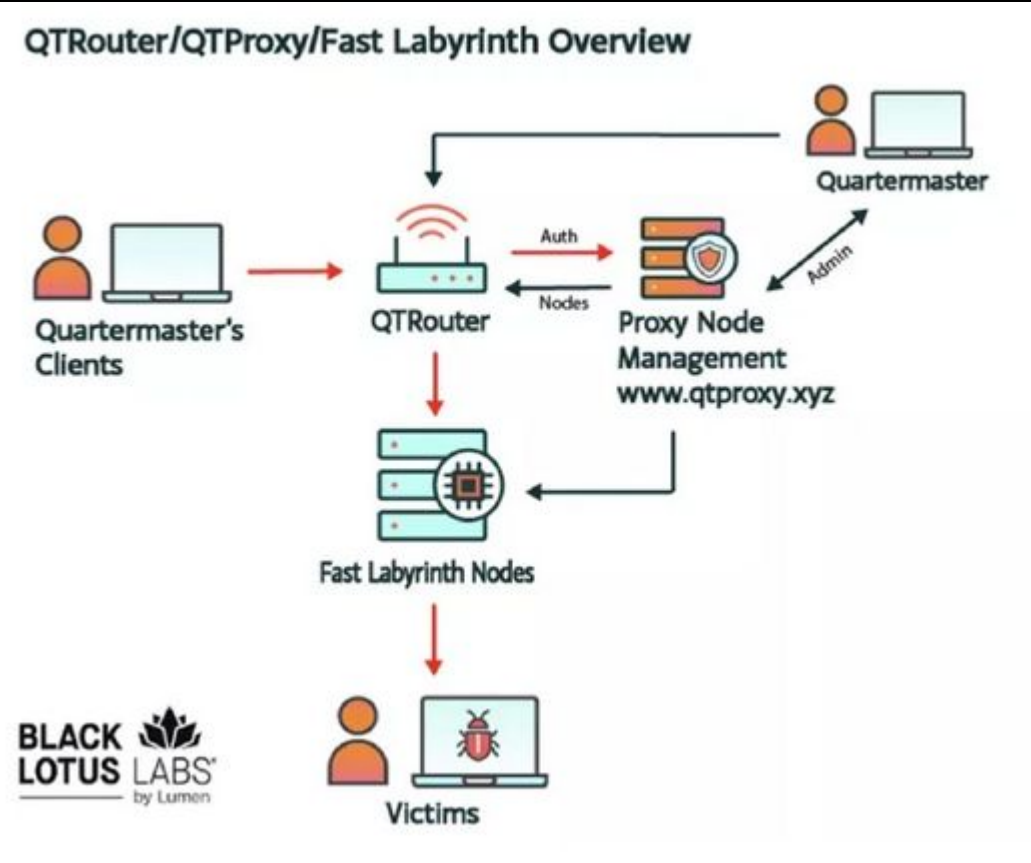
What do we do?

- Learn & train fundamentals
 - Network Security
 - Linux & Windows OS Security
 - Attacking Active Directory
 - Malware Development & Detection
- Competitions
 - DoE CyberForce & CCDC
 - TracerFIRE
- Research
 - Find 0days & write malware



Learn how to do stuff like this!

FBI Disrupts China-Linked QTFY Infrastructure Used to Steal Data From U.S. Organizations



CISA Red Team Compromised Two Critical Infrastructure Orgs, One Detected Nothing

Home > Campaigns > ArcaneDoor

ArcaneDoor

ArcaneDoor is a campaign targeting networking devices from Cisco and other vendors between July 2023 and April 2024, primarily focused on government and critical infrastructure networks. ArcaneDoor is associated with the deployment of the custom backdoors Line Runner and Line Dancer. ArcaneDoor is attributed to a group referred to as UAT4356 or STORM-1849, and is assessed to be a state-sponsored campaign.^{[1][2]}

ID: C0046

First Seen: July 2023^[1]

Last Seen: April 2024^[1]

Contributors: Jun Hirata, NEC Corporation; Sareena Karapoola, NEC Corporation India; Pooja Natarajan, NEC Corporation India

Version: 1.0

Created: 06 January 2025

Last Modified: 31 July 2026

[Version](#) [Permalink](#)

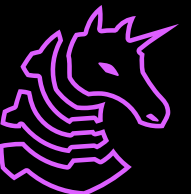
ATT&CK[®] Navigator Layers

Techniques Used

Domain	ID	Name	Use
Enterprise	T1583	.003 Acquire Infrastructure: Virtual Private Server	ArcaneDoor included the use of dedicated, adversary-controlled virtual private servers for command and control. ^[1]
		.006 Acquire Infrastructure: Web Services	ArcaneDoor included the use of OpenConnect VPN Server instances for conducting actions on victim devices. ^[1]
Enterprise	T1557	Adversary-in-the-Middle	ArcaneDoor included interception of HTTP traffic to victim devices to identify and parse command and control information sent to the device. ^[1]
Enterprise	T1071	.001 Application Layer Protocol: Web Protocols	ArcaneDoor command and control activity was conducted through HTTP. ^[1]

We Will Cover

- Basic VM setup
- Networking fundamentals & network security
- OS fundamentals & OS security
- Active Directory authentication & attacks
- How to write your own tools
- Digital Forensics
- Incident Response
- Malware Detection



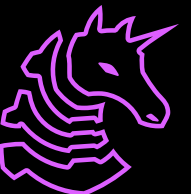
Don't Get Arrested II

- The techniques used here can be used to attack real enterprises
 - If you try to do anything from the first semester, **you will be caught**
 - Do **NOT** try to do this against external targets / UIUC Active Directory
 - The last guy to try this got 24 months in jail and >20k in fines
- We will get into operating undetected later in the semester
 - Operating undetected in modern environments is hard (but not impossible)
 - Do **NOT** try to do this. If you can do this, then you will have no problem finding a high-paying job



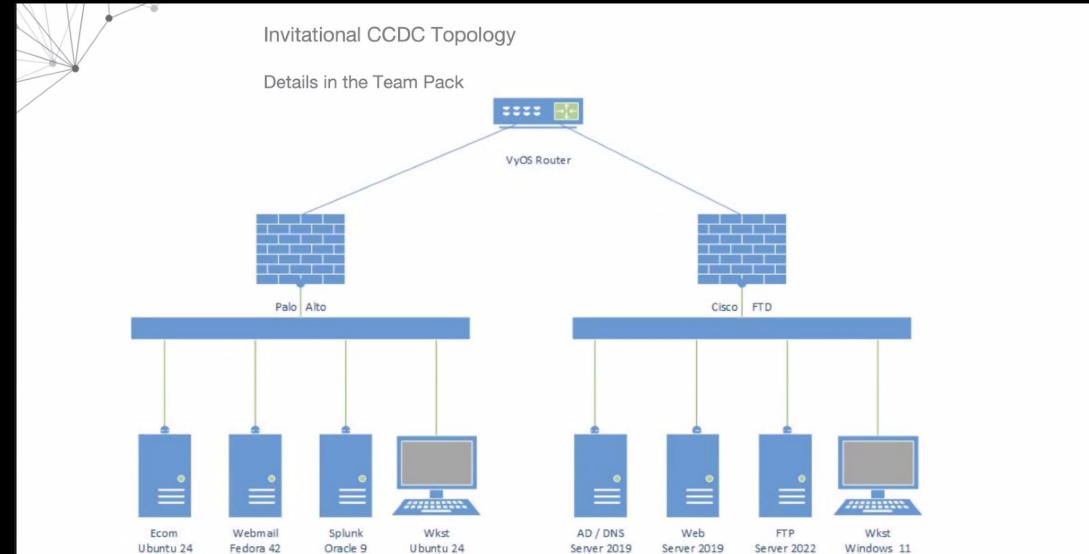
Remote Code Execution != Hacking

- Actual hackers are going to be taking extensive measures to conceal their identity and operation
- Real ethical red teamers must understand their tools
- Being a professional means red teaming responsibly
- This requires attention to detail that will not be explicitly covered in most training
- If you are interested in a **career path** in cybersecurity, feel free to talk to admins and helpers!



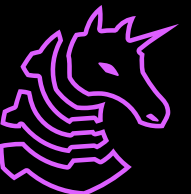
Competitions

- Blue teamers (us) are given systems to secure and maintain the functionality of various services
 - Services are scored by uptime
 - Red team will try to take our services down
- We will be given business tasks to do simultaneously
- Submit Incident Response reports



Competitive Expectations

- Purple represents SIGPwny and UIUC overall externally at collegiate cyberspace events/competitions in cyber defense.
- Some competitions are less restrictive and open-invite. We can bring dozens to participate. Commit availability for this and communicate effectively.
- Some are restricted to one team per university and are the source of our national ranking. We insist on remaining competitive.
 - Roster selections will prioritize those with training/experience, demonstrated trust to the team, and/or meaningful contributions.
 - These environments **will** be tense. The team prefers those with this temperament and aptitude to be considered heavily more.



Build Your Roadmap

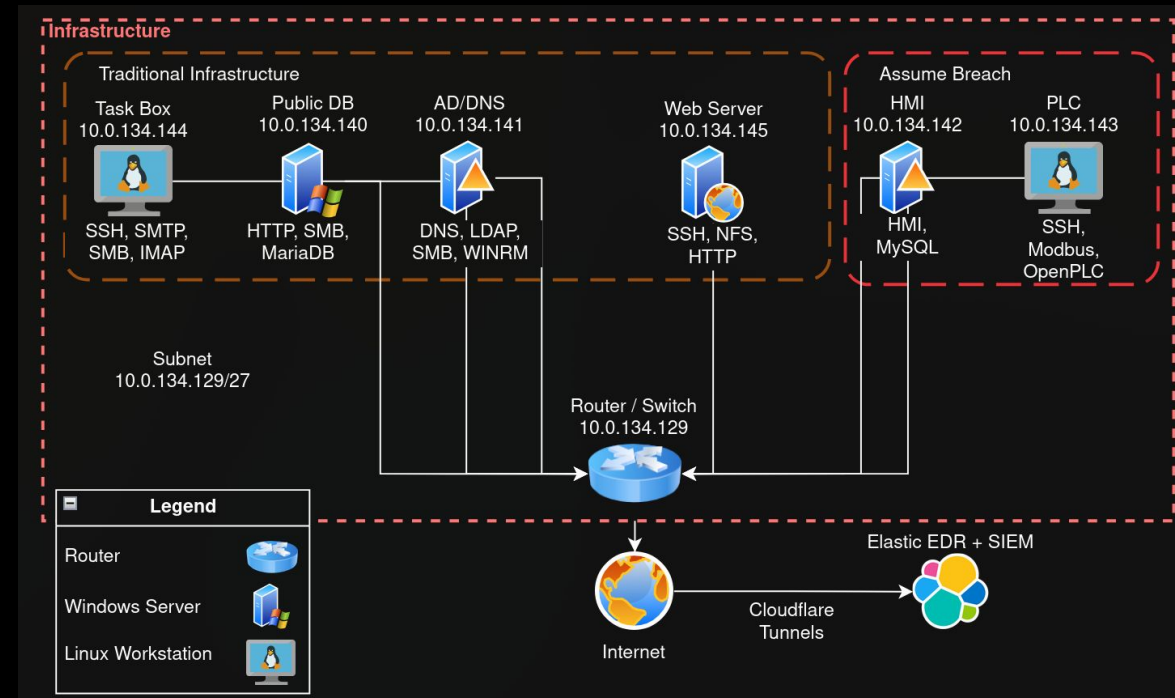
- Cybersecurity is broad, and offensive and defensive involvement span many cyber subdomains. People spend their entire careers pursuing a handful of these domains. You will eventually find what resonates more with you.
- Generally you can begin with defense then pivot to offense
- Then specialize. You can be the SME for that area (subject matter expert). If you compete, you can assume relevant responsibilities on for the team's network. On a day to day, you can contribute with more exposure.



CyberForce

- Defend an **Operational Technology** network
- Last year was an oil rig
- Requires both social and technical skills
- Last year we were the top team for defense and service uptime!

they hated our presentation :(



Anomaly Score	Red Score	Service Score
SIGPwny		T1st
Tinley Park Mental Health Cen...		T1st
CyberFlames		T3rd
pwn@vt		T3rd
BW CyberSec		T5th
CNS@UVA		T5th
Darth Gator		T5th
DayZero		T5th
49th Security Division		T9th
Cyber Spartans		T9th
SeisMICS		T9th



CCDC

- Defend an **Information Technology** network
- Keep a variety of services running in a small simulated enterprise
- You are actively under attack!
- Respond to business tasks (injects)
- Respond to simulated users
- Simulates realistic SOC scenarios under tight and jam-packed deadlines

Service Status: 2026-02-21 13:05:43

	2019-http	2022-http	ad-dns	ecom-http	mail-pop3	mail-smtp	splunk-http
Team 01	↓	↓	↓	↓	↓	↓	↓
Team 02	↓	↓	↓	↓	↓	↓	↓
Team 03	↓	↓	↓	↓	↓	↓	↓
Team 04	↑	↑	↑	↑	↑	↑	↑
Team 05	↓	↓	↓	↓	↓	↓	↓
Team 06	↑	↑	↑	↑	↓	↑	↑
Team 07	↑	↓	↑	↑	↑	↑	↑
Team 08	↑	↑	↑	↑	↑	↑	↑
Team 09	↓	↓	↓	↓	↓	↓	↓
Team 10	↓	↓	↓	↓	↓	↓	↓
Team 11	↑	↑	↑	↑	↑	↑	↑
Team 12	↓	↓	↓	↓	↓	↓	↓
Team 13	↓	↓	↓	↓	↓	↓	↓

University of Illinois at Urbana-Champaign CCDC Team,

It was one of my greatest pleasures to make plans to include your team at the 2026 MWCCDC regional. Your team took 4th place overall, vindicating this decision.

Attached are feedback reports from the recent regional. Note that nearly all teams are assessed 'weak' on services in their report, but U of I was 8th on services. 84% response on injects was very good, and your inject scores were good.

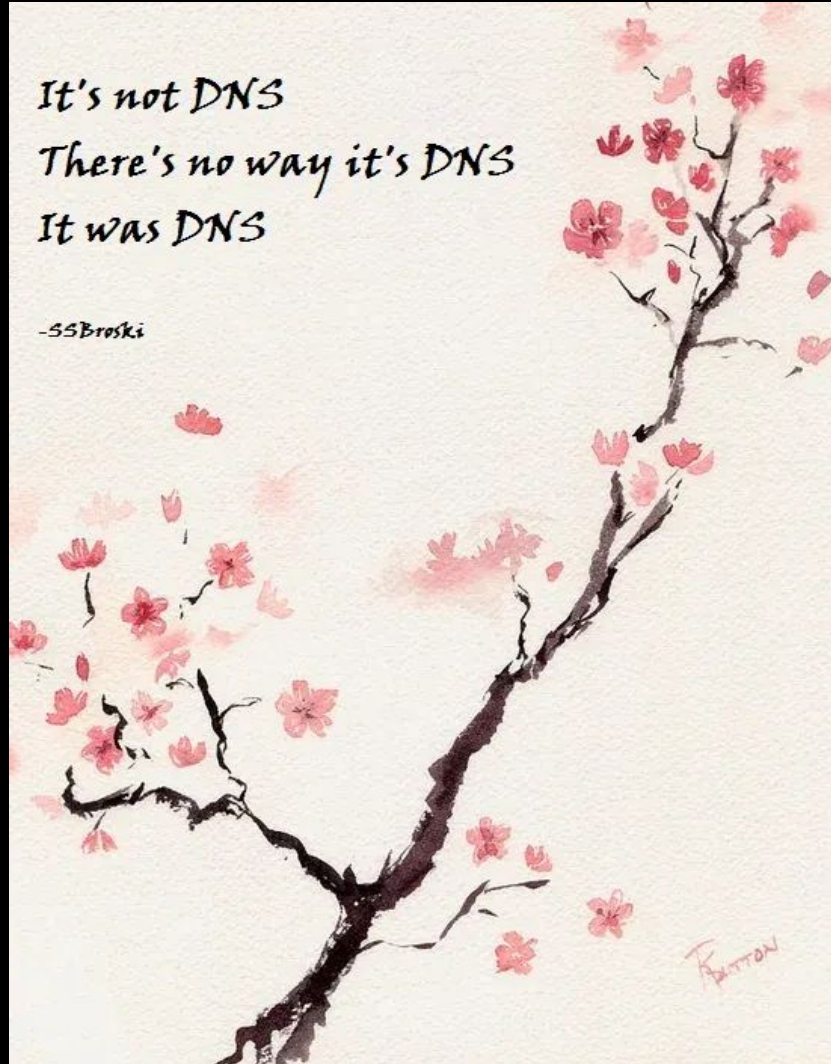
Your assessment from the red team was especially good, 3rd for the event and pretty close to the 2nd place team.

U of I is on the map as one of the very best CCDC teams in the Midwest, and joins the Illinois rivalry with DePaul, SIU, and IIT.

I look forward to working with your team in the 2027 MWCCDC season.

David Durkee

The CCDC Experience



```
=====
Never gonna give you up
Never gonna let you down
Never gonna run around and desert you

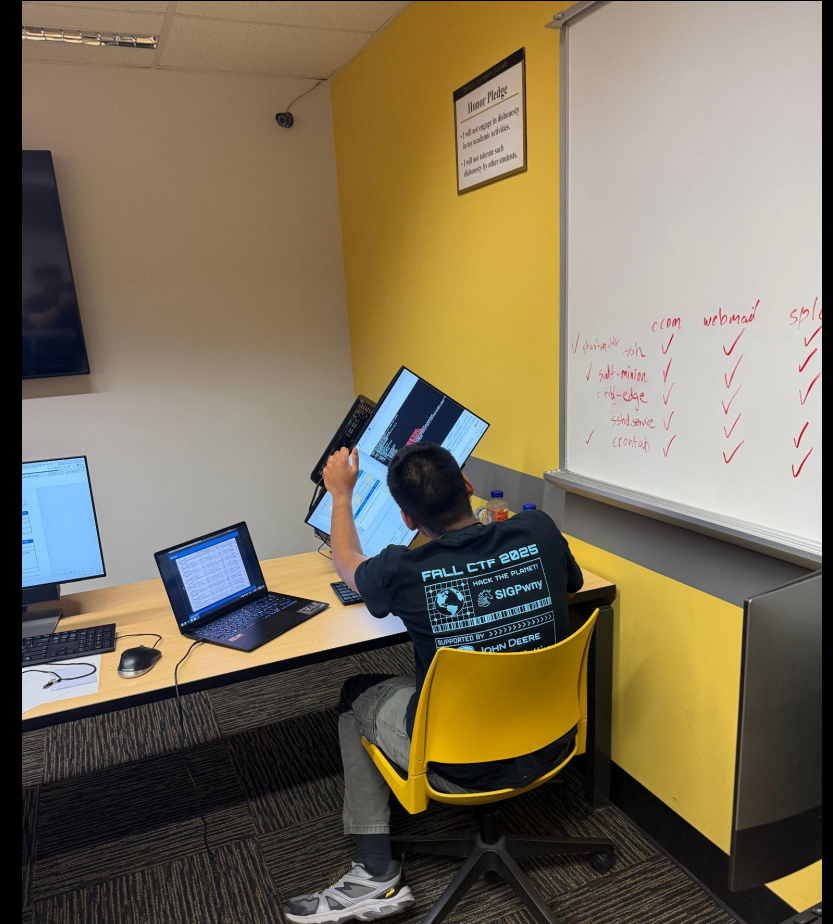
~ Your friendly neighborhood Red Team ~

=====
gc "333w
pb' 2 @l
p @01d
c
_
1a 1 ccglp
#
q
?)a ggr
, a]
p @
d=
hjl= a @egg l=
cc a^ q c]
fa3c =^ b
; , -
db^34-
- a
376
a "37
; c
w "4' @gg
@31 ^jl=
d=pb
' 7= -
>a "3
-3 a
^363
a
=====

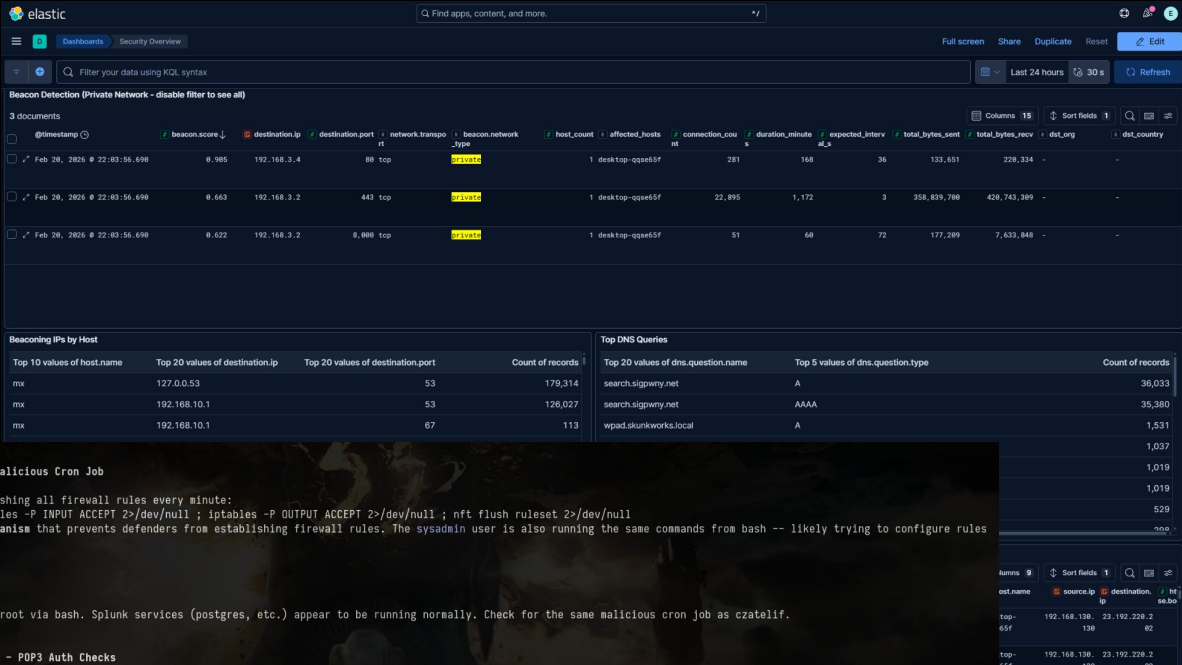
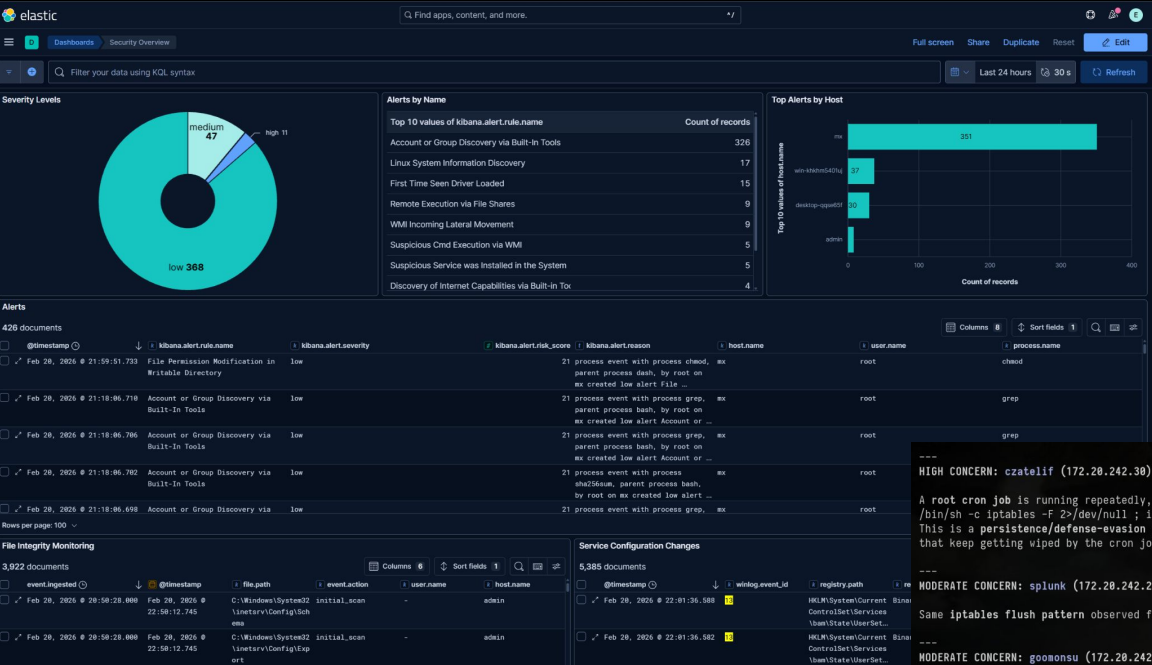
Never gonna give you up
Never gonna let you down
Never gonna run around and desert you

~ Your friendly neighborhood Red Team ~

sysadmin@goomonsu:~$ _
```



Defensive Tool Development



HIGH CONCERN: czatelif (172.28.242.30) - Malicious Cron Job

A root cron job is running repeatedly, flushing all firewall rules every minute:
/bin/sh -c iptables -F 2>/dev/null ; iptables -P INPUT ACCEPT 2>/dev/null ; iptables -P OUTPUT ACCEPT 2>/dev/null ; nft flush ruleset 2>/dev/null
This is a persistence/defense-evasion mechanism that prevents defenders from establishing firewall rules. The sysadmin user is also running the same commands from bash -- likely trying to configure rules that keep getting wiped by the cron job.

MODERATE CONCERN: splunk (172.28.242.20)

Same iptables flush pattern observed from root via bash. Splunk services (postgres, etc.) appear to be running normally. Check for the same malicious cron job as czatelif.

MODERATE CONCERN: goomonsu (172.28.242.40) - POP3 Auth Checks

Running Dovecot POP3 service with repeated unix_chkpwd calls for users like chase.logan and charles.labelle. This could be:
- Legitimate mail service being scored
- Password brute-force attempts against POP3

DOMAINCONTROL-1 (172.28.240.182) - No process telemetry yet

No endpoint process events found in the last 8 hours. It was only registered at 16:01, so this is expected. The DNS beacon to 104.77.144.79:53 (score 0.948) resolves to legitimate Akamai CDN hostnames -- likely benign Windows DNS resolution.

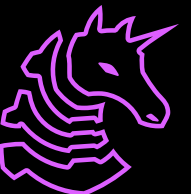
BENIGN / False Positives

Destination	Score	Verdict
67.4.56.154:80 (scoring.ccdscoring.net)	0.665	CCDC scoring engine - all new hosts POSTing to /data. Safe.
1.1.1.1:53 (Cloudflare DNS)	0.834	Normal DNS resolver traffic
172.28.240.102:53 (Domain Controller)	0.832/0.746	Internal DNS to DC. Expected.
104.77.144.79:53 (Akamai DNS)	0.948	Queries for a767.dspw65.akamai.NET etc. Legitimate Akamai CDN lookups
Microsoft IPs (172.183.7.x, 52.x, etc.)	0.7-0.834	Windows telemetry/update traffic
10.48.14.24:8089	0.759	Splunk management port. Likely Splunk forwarder traffic.
192.168.10.53:53	0.558	Internal DNS resolver



TracerFIRE

- Forensics competition, no active defense
- You will have a simulated organization breached by multiple threat groups
- Investigate and figure out:
 - What happened?
 - How did they get in?
 - Who are they?
 - What is their objective?



How To Prepare



Meaningful Training Requires Effort

- Red teaming is **hard**!
- Security is not an academic discipline
- You cannot *learn* things passively
- You must **do** security to actually build proficiency
 - Building even basic proficiency requires repeated practice
 - There are no "experts" in this field - just people with a lot of real experience
- We will provide you an environment to start doing security
 - The internal training can't cover everything, so it's important to find this externally (we will show you where!)



Training Is Encouraged & Noticed Here

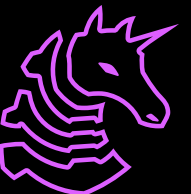
Proficiency in this space is contingent on your ability to employ resources: *walkthroughs, labs, cyber ranges, certifications*

- We strongly encourage you choose pathways within these sites and bring your knowledge to Purple.
 - Start with the fundamentals, then build towards specialization
- Those who do train are evidently more capable in this space and can contribute significantly more
 - This is legitimate resume experience for technical aptitude and training sections. Well respected by employers.
- Training on your own, prior experience, and involvement are the main factors for who makes it onto the most selective teams



Training Resources

- HackTheBox (<https://www.hackthebox.com/>)
 - On-demand vulnerable machines & networks, mostly red team
 - Recently also added general CTF, DFIR, SOC practice as well as structured learning content. **highly recommended**
- PwnyCTF (<https://ctf.sigpwny.com/>)
 - Still a great way to learn basics, less relevant in purple team
- CyberDefenders.org (<https://cyberdefenders.org/>)
 - DFIR & SOC focused with *great* free content
- TryHackMe (<https://tryhackme.com/>)
 - Designed for beginners, good for general training and exposure
 - Free content may be lower quality, updates less frequently than HTB



Certs Are Expensive; Knowledge Is Cheap

- Knowledge
 - Participating in Purple Team - ALL FREE
 - TryHackMe subscription - \$8 per month w/ student discount
 - HackTheBox academy content - \$8 per month w/ student discount
 - HackTheBox all boxes unlock - \$25 per month
 - HackTheBox **advanced pro labs** - \$49 per month
- Certificates
 - Hackthebox CPTS - \$210 per exam attempt
 - eJPT - \$250 per exam
 - PJPT/PNPT - \$500 per exam
 - CRT0 - £365 course + exam attempt
 - Any Offsec cert (OSCP, OSEP, OSED, etc) - **\$1750+**

You don't have to commit to cybersecurity to learn cybersecurity.



Skill Levels

- **Beginner** <- most of you are here
 - able to understand basic programming, OS, networking
 - **willing to engage in structured learning and practice**
- **Intermediate** <- we will cover content until here
 - comfortable performing common offensive and defensive tasks
 - e.g. enumerate and compromise common systems, perform basic privilege escalation and lateral movement, analyze logs and alerts, and investigate straightforward incidents
 - Certificates like CPTS, OSCP, CCD, etc are within reach with targeted prep.
- **Advanced/Career-ready**
 - We won't cover every advanced topic in depth. You should be comfortable exploring independently at this stage.
 - Feel free to talk to our members for guidance



Next Meetings

2026-09-3 • This Thursday

- Setup Meeting! Get a Kali VM up and running

2025-09-16 • Next Tuesday

- Network Security Fundamentals



Training Resources: Deeper Dive

- PwnyCTF (<https://ctf.sigpwny.com/>)
 - Great place to learn the fundamentals, including forensics & reverse engineering!
- CyberDefenders.org (<https://cyberdefenders.org/>)
 - DFIR & SOC focused with *great* free content
- HackTheBox (<https://www.hackthebox.com/>)
 - On-demand vulnerable machines & networks, mostly red team
 - Recently acquired LetsDefend, expanding defensive pathways
- TryHackMe (<https://tryhackme.com/>)
 - Designed for beginners, good for general training and exposure



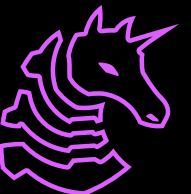
What do I mean by "Red Team"?

- My goal is not to just teach **penetration** testing or security in general, **but to give people the same skill set that real-world threat actors are using to compromise actual organizations**
- We start from the fundamentals and build towards current state-of-the-art techniques
- By the end of the year, you should have a conceptual and practical understanding of what actual bad guys do, how to do it, and the environments they are targeting, and have practical techniques to detect and stop them



Remote Code Execution != Hacking

- Actual hackers are going to be taking extensive measures to conceal their identity and operation
- Real ethical red teamers must understand their tools
- Being a professional means red teaming responsibly
- This requires attention to detail that will not be explicitly covered in most training



Where are we going?

- Starting with basic network & OS security
- The remainder of the semester will cover attacking Active Directory
- I would recommend HackTheBox academy or TryHackMe
- If you have the money and commitment, consider OSCP

If you want a structured learning environment, **HackTheBox Academy** is \$8/month as a student and is ***HIGHLY*** recommended!

